LAB EXPERIMENT 3

Experiment 3: Morphological Analysis Using NLTK

Aim

To perform basic morphological analysis of words using tokenization and lemmatization with the NLTK library.

Algorithm

1. Start the program.
2. Import NLTK's word tokenizer and WordNetLemmatizer.
3. Download the required tokenizer and WordNet data if they are not already installed.
4. Define a sample sentence and tokenize it into words.
5. Use the lemmatizer to find the base form of each word.
6. Display each original word and its lemma.
7. Stop the program.

Code

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer

nltk.download("punkt", quiet=True)
nltk.download("punkt_tab", quiet=True)
nltk.download("wordnet", quiet=True)
nltk.download("omw-1.4", quiet=True)

text = "The cats are playing in the gardens."
words = word_tokenize(text)
lemmatizer = WordNetLemmatizer()

print("Word -> Lemma")
for word in words:
    if word.isalpha():
        lemma = lemmatizer.lemmatize(word, pos="n")
        print(word, "->", lemma)
```

Output / Screenshot

```
import nltk
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()
words = ["playing", "played", "cats", "running"]

for word in words:
    print(word, "->", stemmer.stem(word))
```

```
playing -> play
played -> play
cats -> cat
running -> run
```

Result

Thus, morphological analysis was performed using NLTK by tokenizing the sentence and displaying the base forms (lemmas) of the words.